

Exposé

WAB

LoRaWAN vs NB-IoT: Comparing Unlicensed vs Licensed LPWAN-Technologies

For the module 'Embedded Systems and Software'

submitted to the Department Informatik of
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by

Rubin Chempananickal James
D876
BSc. Informatik BIN23
rubin.chempananickaljames@stud-provadis-hochschule.de
+49 176 27693100

First examiner: Prof. Dr. Eric Hutter

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1. Problem Statement and Motivation

The rapid growth of the Internet of Things (IoT) has led to an increasing demand for Low Power Wide Area Network (LPWAN) technologies that can support a large number of connected devices over long distances. Two prominent LPWAN technologies are Long Range Wide Area Network (LoRaWAN), which operates in unlicensed spectrum, and Narrowband Internet of Things (NB-IoT), which operates in licensed spectrum. This work aims to compare these two technologies in terms of their performance, advantages, and disadvantages in various application scenarios. The motivation for this comparison is to provide insights into which technology may be more suitable for specific use cases and to contribute to the ongoing discussion about the future of IoT connectivity.

2. Objectives

The main objective of this work is to compare the two LPWAN technologies, LoRaWAN and NB-IoT, in terms of their performance, advantages, and disadvantages in various application scenarios. The specific objectives include:

3. Research Questions

- What are the differences between LoRaWAN and NB-IoT in terms of range, data rate, power consumption, and cost?
- How do LoRaWAN and NB-IoT perform in real-world scenarios, such as urban and rural environments?

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- What are the advantages and disadvantages of using unlicensed (LoRaWAN) vs licensed (NB-IoT) spectrum for LPWAN technologies?

4. State of the Art

The state of the art in LPWAN technologies includes various studies and comparisons of LoRaWAN and NB-IoT. LoRaWAN is known for its long-range capabilities and low power consumption, making it suitable for applications such as smart agriculture and environmental monitoring. NB-IoT, on the other hand, offers higher data rates and better reliability, which can be advantageous for applications like smart cities and industrial IoT. However, there is still a need for comprehensive comparisons that consider real-world performance and the implications of using unlicensed vs licensed spectrum.

5. Methodology

The methodology for this work will involve a comprehensive literature review of existing studies on LoRaWAN and NB-IoT, followed by experimental evaluations in different environments to assess their performance. Data will be collected on key metrics such as range, data rate, power consumption, and cost. The results will be analyzed to identify the strengths and weaknesses of each technology in various application scenarios.

6. Planned Structure

1. Introduction
2. State of the Art / Theory
3. Methodology
4. Investigation / Implementation
5. Results
6. Conclusion and Outlook

7. Glossary and Abbreviations

7.1 Glossary

7.2 Abbreviations

8. References

Bibliography

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